Anesthetic Gases

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Continuing Education Activity

Anesthetic gases (nitrous oxide, halothane, isoflurane, desflurane, sevoflurane), also known as inhaled anesthetics, are administered as primary therapy for preoperative sedation and adjunctive anesthesia maintenance to intravenous (IV) anesthetic agents (i.e., midazolam, propofol) in the perioperative setting. Inhaled anesthetics enjoy regular use in the clinical setting due to chemical properties that allow the rapid introduction of an agent into arterial blood via the pulmonary circulation compared to the more circuitous route of venous circulation. This activity reviews the mechanism of action, adverse event profile, and other key factors (e.g., dosing, pharmacodynamics, pharmacokinetics, monitoring, relevant interactions) pertinent to members of the interprofessional team for the use of these anesthetic gases.

Objectives:

- Identify the approved and off-label uses for anesthetic gases.
- Summarize the general mechanism of action for anesthetic gases.
- Review the potential adverse events and current research on contraindications with the use of anesthetic gases.
- Outline interprofessional team strategies for improving care coordination and communication to advance appropriate clinical outcomes when using anesthetic gases where applicable.

Access free multiple choice questions on this topic.

Indications

Anesthetic gases (nitrous oxide, halothane, isoflurane, desflurane, sevoflurane), also known as inhaled anesthetics, are administered as primary therapy for preoperative sedation and adjunctive anesthesia maintenance to intravenous (IV) anesthetic agents (i.e., midazolam, propofol) in the perioperative setting.[1] Inhaled anesthetics enjoy regular use in the clinical setting due to chemical properties that allow the rapid introduction of an agent into arterial blood via the pulmonary circulation compared to the more circuitous route of venous circulation.[2] The significance of rapid therapeutic effects allows for efficient induction and discontinuation of sedation induced by these agents, providing proper amnesia, anesthesia, and a faster recovery period in postoperative care than IV agents.[3]

Though indicated solely for the perioperative setting, these agents also have a significant off-label use within critical care to facilitate patient tolerance of endotracheal intubation, mechanical ventilation, and bedside procedures. Generally, for these cases, the recommended use of IV benzodiazepines (midazolam, lorazepam, diazepam) or propofol induces this level of sedation.[4] However, more recent studies have explored the regular use of inhaled anesthetics, specifically the volatile anesthetics (halothane, isoflurane, desflurane, sevoflurane), as first-line agents for critical care sedation. Preliminary findings show shorter times to extubation and shorter lengths of stay in the ICU; however, there is a need for further study of these agents in this setting.[2][3][4]

FDA-label Indications

- Preoperative sedation – primary or adjunctive; rapid induction of sedation, providing amnesia and anesthesia during surgical procedures.
- Perioperative sedation maintenance – adjunctive; maintains anesthesia after sedation by IV benzodiazepines or propofol.

FDA-off-label Indications

- ICU sedation – primary or adjunctive; facilitates tolerance of intubation, mechanical ventilation, bedside procedures.

Mechanism of Action

Inhaled anesthetics work to depress neurotransmission of excitatory paths involving acetylcholine (muscarinic and nicotinic receptors), glutamate (NMDA receptors), and serotonin (5-HT receptors) within the central nervous system (CNS) and augment inhibitory signals including chloride channels (GABA receptors) and potassium channels to provide an adequate level of sedation.[5] These agents are sub-classified by both their chemical properties and believed mechanisms of action:

- **Non-volatile gases:** nitrous oxide (N2O)
- **Volatile gases:** halothane, isoflurane, desflurane, sevoflurane

The main distinction between the non-volatile and volatile gases originally stemmed from their specific chemical properties. Non-volatile anesthetics have high vapor pressures and low boiling points, meaning they are in gas form at room temperature, whereas volatile anesthetics have low vapor pressures and high boiling points, meaning they are liquids at room temperature and so require vaporizers during administration.[6] Since these agents work on many different receptors, as described above, physiologically distinguishing these subclasses has proved to be more arduous. Current thought suggests non-volatile agents primarily inhibit NMDA receptors and glutamate signaling, whereas volatile agents augment GABA signaling.[5]

Administration

Administration of all anesthetic gases is via inhalation. As described above, N2O being the only non-volatile gas clinically administered at room temperature in its gaseous state, whereas the volatile gases of halothane, isoflurane, desflurane, and sevoflurane are liquids at room temperature requiring a vaporizer for administration.[6] Compared to other agents in pharmacology, where the basis of the therapeutic index is the bioavailability of the agent within serum determined via the route of administration (IV, PO, IM, SC), inhaled anesthetics are unique in that they have one route of administration and multiple factors, listed below, that determine therapeutic index[7]:

- Minimum alveolar concentration (MAC): used as a measure of potency, defined as the % gas concentration determined to produce immobility to noxious stimuli in 50% of patients. Essentially, the higher the MAC, the less the potency of the gas for sedative purposes.
- Alveolar concentration (FA) to inspired concentration (FI): used to determine the speed of induction, works on a 0 to 1 scale where the FA/FI will approach 1 as the gas gets administered, 1 being equilibrium. The rate at which the above ratio approaches equilibrium represents the speed of induction.
- Partition Coefficients: used to determine the solubility of the gas in both arterial blood (blood to gas) and perfused tissues (brain to blood). These coefficients represent the amount of gas that can enter the blood and the amount that must be reversed for equilibrium to be achieved, forming an inverse relationship with the speed of induction (FA/FI). The higher the partition coefficient, the higher concentration that can enter the blood or the brain, and the more that must flow backward along that gradient to reach equilibrium, which causes a slower rate of induction.

All agents have individualized aspects concerning administration[7]:

- **N2O:** greater than 100% MAC, blood to gas partition coefficient of 0.47, brain to blood partition coefficient of 1.1
- **Halothane:** 0.75% MAC, blood to gas partition coefficient of 2.30, brain to blood partition coefficient of 2.9
- **Isoflurane:** 1.4% MAC, blood to gas partition coefficient of 1.4, brain to blood partition coefficient of 2.6
- **Desflurane:** 6 to 7% MAC, blood:gas partition coefficient of 0.42, brain:blood partition coefficient of 1.3
- **Sevoflurane:** 2.0% MAC, blood:gas partition coefficient of 0.69, brain:blood partition coefficient of 1.7

Adverse Effects

Though relatively benign for immediate adverse reactions, anesthetic gases have been observed in specific individuals to cause malignant hyperthermia. Classically, halothane was the most common agent causing this reaction; however, all volatile gases (halothane, isoflurane, desflurane, sevoflurane) and depolarizing neuromuscular blockers (succinylcholine) have induced this reaction.[8] Patients susceptible to this reaction possess heritable alterations within various proteins involved in the modulation of muscular cytosolic concentrations of Ca2+. The most common modifications found include the ryanodine receptor/channel encoded by gene RYR1 and dihydropyridine receptor (DPHR) encoded by gene CACNA1S. Both sarcoplasmic proteins are involved with Ca2+ transport, so in the event of their alteration and subsequent exposure to the volatile gases or succinylcholine, excessive release of Ca2+ in skeletal muscle results.[9] Potential symptoms from this hypermetabolic process include muscle rigidity, hyperthermia, rapid onset tachycardia, hypercapnia, hyperkalemia, metabolic acidosis.[5] To quickly reverse this process, the administration of dantrolene is a must; the mechanism of action is the reduction of Ca2+ release from the sarcoplasmic reticulum. Besides administering dantrolene, appropriate measures to reduce body temperature and restore electrolyte and acid-base imbalances are recommended.[8]

There has been a suggestion that these agents are involved with postoperative nausea and vomiting (PONV). There is only a confirmed correlation between the administration of general anesthesia, both IV and inhalation agents, and PONV incidence.[10] Several studies suggest an association between specifically N2O administration and PONV incidence; however, others have contradicted these findings.[10][11][12] Independent of the cause, to correct this reaction, usually anti-emetic agents (ondansetron, metoclopramide, dexamethasone, etc.) are administered prophylactically and can be administered postoperatively as “rescue” agents.[7]

Contraindications

Anesthetic gases, specifically volatile gases, have generalized contraindications in patients with known susceptibility to malignant hyperthermia as described above and use in patients with severe hypovolemia and intracranial hypertension due to the negative inotropic effects and effect of increasing cerebral blood flow that the gases can cause.[7][8] There are contraindications worth mentioning that are more drug specific[5]:

- **N2O**—venous or arterial air embolus, pneumothorax, acute intestinal obstruction with bowel distention, pneumocephalus following dural closure, pulmonary air cysts, intraocular air bubbles, tympanic membrane grafting
- **Halothane**—unexplained liver dysfunction following previous anesthetic exposure, reduced ejection fraction heart failure, or pheochromocytoma
- **Isoflurane**—relative contraindication in patients with severe asthma or active bronchospasm due to the pungency of the agent
- **Desflurane**—relative contraindication in patients with severe asthma or active bronchospasm due to the pungency of the agent
- **Sevoflurane**—relative contraindication in patients presenting with renal dysfunction undergoing extensive surgical procedures

Monitoring

Several key patient parameters require monitoring when using anesthetic gases to induce and maintain general anesthesia during the perioperative period. First: the patient’s vital signs due to the suppressive effects on the CNS and sympathetic nervous system that these agents exhibit. Heart rate (HR) and blood pressure (BP) specifically are monitored by the minute to detect tachyarrhythmias or sudden hypertension/hypotension.[7] Temperature is critical to monitor to detect early signs of malignant hyperthermia.[8] Third, one must ensure and monitor a secure airway for continued delivery and manipulation of anesthetic gases throughout the procedure. Ventilation can be important in the delivery of these gases and the reversal of their effects. Achieving these goals involves monitoring and manipulating the end-tidal CO2, tidal volume, and respiratory rate via mechanical ventilation.[13] Other organ systems which require continued monitoring and possible correction are[7]:

- Cerebral effects: nitrous oxide can increase cerebral blood flow and cause increased intracranial pressure, whereas volatile anesthetics can decrease cerebral metabolic rate, decreasing cerebral blood flow to a point until reaching 1.5 MAC, and resulting vasodilation can occur, increasing the intracranial pressure. Both situations ameliorate with the use of IV anesthetics and/or hyperventilation, which reduces cerebral blood flow and decreases intracranial pressure.
- Cardiovascular effects: all anesthetic gases have shown a depression of cardiac contractility; halothane reduces arterial pressure, and the other volatile gases decrease vascular resistance due to vasodilatation resulting in preserved cardiac output. These effects can cause sympathetic activation and reflex tachycardia, especially with the use of isoflurane and desflurane, which cause less baroreceptor reflex depression than the others. Volatile gases can also sensitize the myocardium to catecholamines, potentially increasing the risk of ventricular dysrhythmias if patients were given sympathomimetic drugs perioperatively. These effects can be balanced with BP (hydralazine, labetalol, verapamil, etc.) and HR control (metoprolol, amiodarone, etc.).
- Pulmonary effects: besides N2O, all anesthetic gases cause a dose-dependent decrease in tidal volume and an increase in respiratory rate resulting in a shallow breathing pattern. All gases are respiratory depressants, as the reduction of ventilatory response to increasing blood CO2 occurs. Mechanical ventilation is required to compensate for this response.

Bispectral index (BIS) is commonly applied with the use of anesthetic gases for general anesthesia to determine the level of sedation achieved. This monitoring technique detects brain activity, similarly to EEG monitoring, and gives an average value on a numerical scale (0 to 100), with values below 60 determined as the level of minimal sedation and below 40 as the level of deep sedation. Though still used commonly in clinical practice, future use of BIS for this type of monitoring is in question because of the lack of studies and no proven gold standard for comparison.[14]

Toxicity

As stated above, these gases are relatively benign for acute adverse reactions.[9] The toxic profiles for these gases fall into acute and chronic toxicities.

Acute Toxicities [15]

- Nephrotoxicity – can occur with all volatile gases. Sevoflurane is metabolized by CYP 2E1 at a much faster rate compared to the other volatile gases and produces increased levels of inorganic fluoride (F), which has been shown during *in vivo* studies of rats to cause renal impairment. Clinically this renal impairment has not been reported; however, the recommendation is still that sevoflurane not be used in patients with preexisting renal dysfunction undergoing extensive procedures.
- Hepatotoxicity – uncommon, but reports exist of individuals with prior exposure specifically to halothane. Patients would present immediately after a procedure with halothane-induced fulminant hepatic failure.
- Carbon monoxide (CO) poisoning – is a concern as all inhaled anesthetics can produce CO from interaction with the dry CO2 absorbers used in the perioperative setting. The largest producer of CO in this manner is desflurane. This situation can is avoidable by routinely changing the absorbent before each day of cases.

Chronic Toxicities [15]

- Hematotoxicity – chronic exposure to N2O has shown to decrease methionine synthase activity, an important enzyme in the recycling of vitamin B. Prolonged vitamin B deficiency can result in megaloblastic anemia, and peripheral neuropathies in patients who work in dental practices should prompt clinical suspicion of this etiology.
- Teratogenic/reproductive effects – early exposure can cause cognitive impairment later in life, particularly N2O, as described above, which can disrupt vitamin B-dependent metabolic pathways.
- Carcinogenic – continued exposure to trace concentrations of anesthetic gases by operating room personnel correlated with increasing rates of cancer diagnosis in these individuals

Overdose or Postoperative Reversal

There is no pharmacological reversal of anesthetic gases in use for postoperative recovery or in the event of an overdose. The primary reversal method is to remove the patient from continued exposure to gas and hyperventilate to decrease the concentration of gas in the patient’s alveoli.[7]

Enhancing Healthcare Team Outcomes

Anesthetic gas use is chiefly the purview of the anesthesiologist, anesthesia nurse, and emergency department physician. However, to ensure the patient's safety, the practice of anesthesia is best done with an interprofessional team that also includes anesthesia nurses, recovery room nurses, and anesthesia assistants. Since there is no antidote to an anesthetic gas, the patient requires close monitoring during anesthesia. A dedicated nurse should monitor the vital signs continuously during induction and maintenance to ensure that the patient is not only sedated but has no pain. There are standard protocols for patient monitoring during anesthesia established by all hospitals.

Pharmacy is responsible for the delivery of the proper agents and ensuring that dosing and administration are correct, providing a "second set of eyes" to assist the anesthesiologist or nurse anesthetist.

After completing the procedure, the patient must be monitored by the recovery room nurses until the patient is fully awake and hemodynamically stable. The anesthesiologist should receive immediate notification if there are problems with the vital signs, oxygenation, and respiration in the recovery room. More crucially, resuscitation equipment should always be available when using anesthetic gases. Ultimately, the anesthesia nurse is responsible for ensuring that the gas canisters are full and that all the monitoring equipment is in working order. Any deficits require the attention and input of the entire interprofessional healthcare team; this is the only way to ensure patient safety. [Level 5]